2008年真题参考答案

一、选择题

(1)B. (2)C. (3)B. (4)A. (5)C. (6)D. (7)A. (8)D.

二、填空题

(9)1. (10)$\frac{1}{2}\ln 3$. (11)$\frac{\pi}{4}$. (12)$\frac{1}{x}$. (13)3. (14)$\frac{1}{2e}$.

三、解答题

(15)$-\frac{1}{6}$.

(16)(Ⅰ)$dz=\frac{2x-\varphi'(x+y+z)}{\varphi'(x+y+z)+1}dx+\frac{2y-\varphi'(x+y+z)}{\varphi'(x+y+z)+1}dy$.

(Ⅱ)$\frac{\partial u}{\partial x}=\frac{-2(2x+1)\varphi''(x+y+z)}{[\varphi'(x+y+z)+1]^3}$.

(17)$\frac{19}{4}+\ln 2$.

(18)证明略.

(19)A至少应为3 980万元.

(20)(Ⅰ)证明略.

(Ⅱ)$a\neq 0$时,方程组有唯一解,$x_1=\frac{n}{(n+1)a}$.

(Ⅲ)$a=0$时,方程组有无穷多解,通解为$\boldsymbol{x}=k(1,0,\cdots,0)^{\mathrm{T}}+(0,1,0,\cdots,0)^{\mathrm{T}}$,其中$k$为任意常数.

(21)(Ⅰ)证明略.

(Ⅱ)$\begin{pmatrix}-1&0&0\\0&1&1\\0&0&1\end{pmatrix}$.

(22)(Ⅰ)$\frac{1}{2}$.

(Ⅱ)$f_Z(z)=\begin{cases}\frac{1}{3}, & -1\leqslant z<2,\\ 0, & \text{其他}.\end{cases}$

(23)(Ⅰ)证明略.

(Ⅱ)$\frac{2}{n(n-1)}$.